Causal Phase Recognition for Wall-Normal Shared Control in Remote Robotic Painting

I. Introduction

- Motivation
  - Observation from the preceding remote-painting experiment
    - Persistent difficulty maintaining detailed EOAT–wall contact during teleoperation
    - Concurrent operator demand: tangential path control and wall-normal contact correction
  - Shared-control opportunity
    - Operator authority over tangential painting motion
    - Robot assistance for the wall-normal component and contact regulation
- Need for causal phase recognition
  - Phase-dependent assistance logic
    - FREE MOTION: assistance disengagement
    - APPROACH: safe convergence without premature contact locking
    - PAINTING: wall-normal regulation during surface interaction
    - RETRACT: prompt assistance release without resisting withdrawal
  - No fixed phase order; no imposed minimum phase duration
  - Operational risk from delayed, missed, or spurious phase transitions
- Research problem
  - Rule-based logic versus data-driven causal classifiers under a common input and replay protocol
  - Dominant PAINTING class versus brief APPROACH and RETRACT intervals
  - Generalization across training-set composition, trials, and participants
- Study aim
  - Comparative evaluation of deterministic, framewise, probabilistic sequential, and recurrent models
  - Effect of four-trial training-set composition and participant-independent generalization
- Research questions
  - RQ1. Which causal model offers the most useful trade-off among phase F1, transition behavior, and interpretability?
  - RQ2. How do four-trial composition and participant holdout alter causal classification performance?

II. Study Data and Phase Definition

- Source dataset
  - Preceding remote robotic painting experiment
  - Six participants; four visual-interface conditions; 22 trials; 61 continuous sequences
  - 208,430 labeled frames; 207,488 frames with valid causal features; 115.8 min
- Causal input representation
  - 12 dimensions per frame
    - TCP force: x, y, z
    - Position relative to trial start: x, y, z
    - Backward-difference velocity: x, y, z
    - Backward-difference acceleration: x, y, z

    - Training-set standardization only; no future-frame features
- Manual phase labels
    - FREE MOTION, APPROACH, PAINTING, RETRACT
    - Dataset distribution
        - FREE MOTION: 29.6%; APPROACH: 5.0%; PAINTING: 62.7%; RETRACT: 2.8%
    - Minority-phase emphasis: APPROACH and RETRACT

III. Causal Phase-Recognition Methods

- Causal FSM
    - Training-calibrated force and motion thresholds; deterministic online state logic
- Random Forest
    - Nonlinear framewise tree ensemble; independent prediction per observation
- Gaussian Mixture Model
    - Phase-conditional mixture likelihoods; framewise maximum-posterior prediction
- Gaussian HMM
    - Single-Gaussian state emissions; causal forward filtering
- GMM-HMM
    - Mixture emissions with learned state transitions; causal forward filtering
- Causal LSTM
    - One-layer recurrent classifier; 64 hidden units
    - 120-frame training chunks; batch size 32; 20 epochs; learning rate 0.001
    - Unweighted cross-entropy; persistent hidden state during one-frame online replay

IV. Evaluation Protocol

- Training-structure comparison; common held-out trial 08_01
    - Participant-specific: 10_01–10_04; four trials
    - Single-interface: 06_01, 07_01, 10_01, 11_01; four trials
    - Diversity-balanced: 06_03, 09_04, 10_01, 11_02; four trials
- Participant-independent evaluation
    - Leave-one-participant-out; six held-out participant folds
- Online replay
    - One observation per model step; temporal state reset only at sequence boundaries
    - No future frames; no test-trial scaler fitting; no test-trial model fitting
- Metrics
    - Frame classification: ±1.0-s temporal-tolerance accuracy; strict full-sequence Macro F1 and per-phase F1
    - Transition detection: precision, recall, and F1 for the same from-to phase event matched one-to-one within ±1.0 s
    - Operational stability: false transitions, absolute transition delay, inference time
    - Tolerance rule: a phase prediction is accepted by tolerant accuracy when the same manual phase occurs within ±1.0 s in the same contiguous sequence; full-sequence F1 remains exact frame-by-frame

V. Results

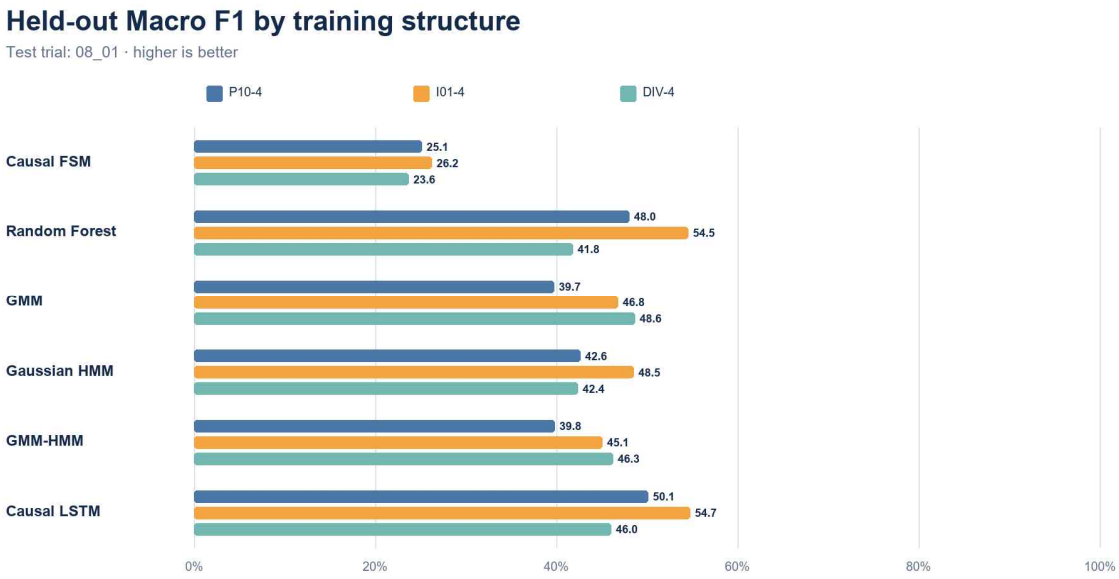
- Four-trial training-set composition (Tables 1–4; Figures 1–4)
    - With four training trials, model ranking depended on how the trials were selected; no four-trial structure dominated every model.
    - Random Forest remained highly competitive in low-data settings, indicating a lower sample requirement than the recurrent model.

Table 1
Held-Out Classification Performance by Training-Set Structure

Model	P10-4	I01-4	DIV-4
Causal FSM	67.1 / 25.1	45.8 / 26.2	54.4 / 23.6
Random Forest	88.3 / 48.0	88.9 / 54.5	87.5 / 41.8
GMM	80.3 / 39.7	83.4 / 46.8	82.7 / 48.6
Gaussian HMM	84.1 / 42.6	83.9 / 48.5	81.1 / 42.4
GMM-HMM	78.8 / 39.8	78.0 / 45.1	79.5 / 46.3
Causal LSTM	83.6 / 50.1	86.2 / 54.7	81.1 / 46.0

Note. Values are ±1-s tolerant accuracy / strict full-sequence Macro F1 in percentages. All models were evaluated on the same unseen trial, 08_01. P10-4 = four trials from participant 10; I01-4 = four trials from interface condition 01; DIV-4 = four condition-diverse trials.

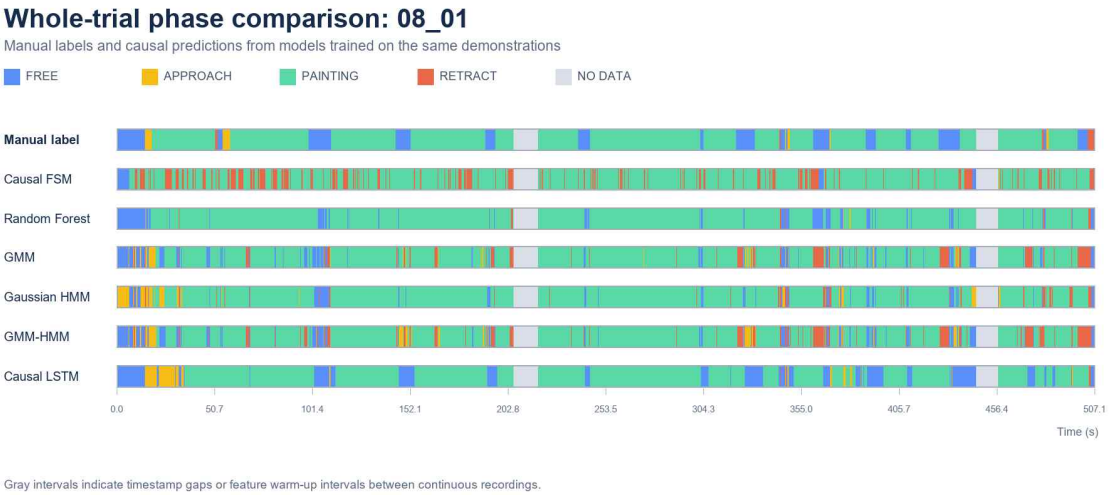
Figure 1
Macro F1 Across Three Four-Trial Training Sets



Note. Model ranking changed with participant and interface composition even though each training set contained four trials.

Figure 2

Whole-Trial Phase Labels for Participant-specific, 4 trials



Note. Ground truth and one-frame causal predictions on 08_01. Training set: 10_01–10_04; 27,813 training frames. Gray intervals indicate timestamp gaps or feature warm-up intervals.

Table 2

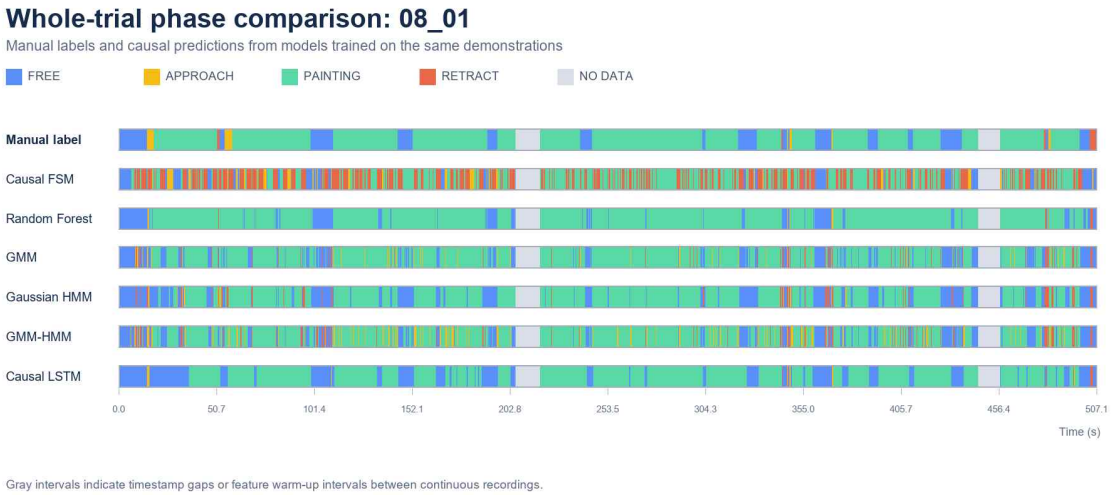
Detailed Performance for Participant-Specific Four-Trial Training

Model	±1 s Tol. Acc.	Full-seq. Macro F1	FREE F1	APP. F1	PAINT. F1	RET. F1	Trans. F1 (±1 s)	Trans. recall	False trans.	|Delay| ms
Causal FSM	67.1	25.1	16.9	0.0	77.3	6.2	1.1	4.9	318	272
Random Forest	88.3	48.0	53.4	11.4	90.9	36.4	11.9	48.8	276	326
GMM	80.3	39.7	43.1	16.7	90.0	9.1	5.6	39.0	510	321
Gaussian HMM	84.1	42.6	43.0	14.0	89.7	23.8	13.3	39.0	183	444
GMM-HMM	78.8	39.8	44.7	14.5	90.6	9.4	6.5	26.8	287	470
Causal LSTM	83.6	50.1	60.3	25.0	86.3	28.9	19.4	29.3	71	512

Note. Results on unseen trial 08_01 after participant-specific four-trial training. Tolerant accuracy accepts phase timing differences within ±1 s; Macro and phase F1 use strict matching over all valid frames. Transition F1 matches the same from-to event within ±1 s. False transitions are unmatched predicted events; delay is in milliseconds. Training set: 10_01-10_04; 27,813 training frames.

Figure 3

Whole-Trial Phase Labels for Single-interface, 4 trials



Note. Ground truth and one-frame causal predictions on 08_01. Training set: 06_01, 07_01, 10_01, 11_01; 33,251 training frames. Gray intervals indicate timestamp gaps or feature warm-up intervals.

Table 3

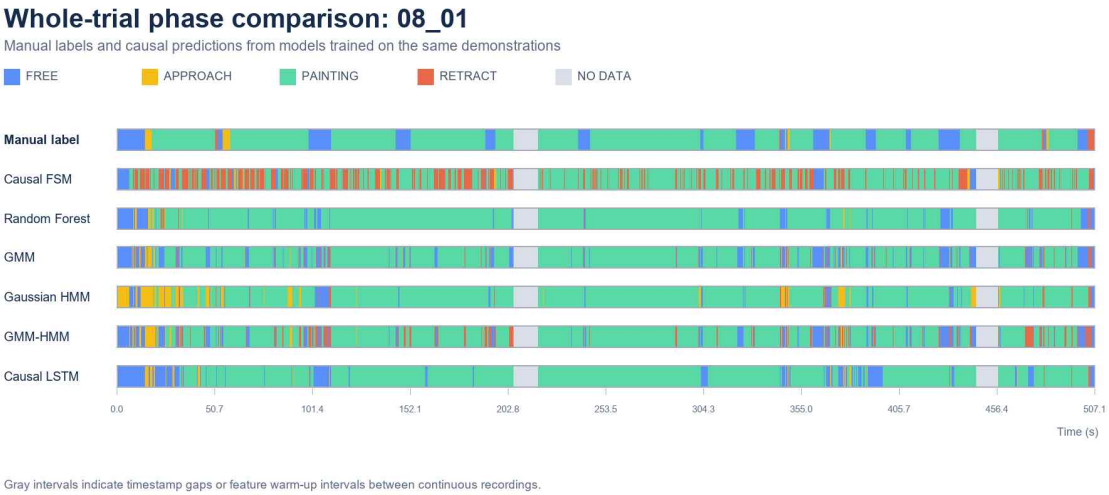
Detailed Performance for Single-Interface Four-Trial Training

Model	±1 s Tol. Acc.	Full-seq. Macro F1	FREE F1	APP. F1	PAINT. F1	RET. F1	Trans. F1 (±1 s)	Trans. recall	False trans.	|Delay| ms
Causal FSM	45.8	26.2	39.3	3.8	60.2	1.6	2.3	17.1	560	260
Random Forest	88.9	54.5	60.7	28.4	90.8	38.1	17.0	58.5	218	329
GMM	83.4	46.8	64.5	15.1	88.6	18.8	7.4	75.6	771	231
Gaussian HMM	83.9	48.5	69.2	13.5	89.3	22.0	15.2	51.2	215	260
GMM-HMM	78.0	45.1	64.5	16.9	85.2	13.7	8.2	61.0	547	276
Causal LSTM	86.2	54.7	66.0	30.9	89.0	33.1	32.6	53.7	72	415

Note. Results on unseen trial 08_01 after single-interface four-trial training. Tolerant accuracy accepts phase timing differences within ±1 s; Macro and phase F1 use strict matching over all valid frames. Transition F1 matches the same from-to event within ±1 s. False transitions are unmatched predicted events; delay is in milliseconds. Training set: 06_01, 07_01, 10_01, 11_01; 33,251 training frames.

Figure 4

Whole-Trial Phase Labels for Diversity-balanced, 4 trials



Note. Ground truth and one-frame causal predictions on 08_01. Training set: 06_03, 09_04, 10_01, 11_02; 27,451 training frames. Gray intervals indicate timestamp gaps or feature warm-up intervals.

Table 4

Detailed Performance for Diversity-Balanced Four-Trial Training

Model	±1 s Tol. Acc.	Full-seq. Macro F1	FREE F1	APP. F1	PAINT. F1	RET. F1	Trans. F1 (±1 s)	Trans. recall	False trans.	|Delay| ms
Causal FSM	54.4	23.6	20.5	1.6	68.2	4.3	0.7	4.9	527	146
Random Forest	87.5	41.8	50.3	10.8	89.9	16.3	9.7	36.6	252	588
GMM	82.7	48.6	55.0	35.3	88.6	15.6	8.2	58.5	522	312
Gaussian HMM	81.1	42.4	35.2	16.7	87.6	30.0	13.8	34.1	148	438
GMM-HMM	79.5	46.3	56.0	32.7	88.1	8.3	9.3	43.9	330	260
Causal LSTM	81.1	46.0	40.7	33.1	85.6	24.6	11.5	22.0	107	360

Note. Results on unseen trial 08_01 after diversity-balanced four-trial training. Tolerant accuracy accepts phase timing differences within ±1 s; Macro and phase F1 use strict matching over all valid frames. Transition F1 matches the same from-to event within ±1 s. False transitions are unmatched predicted events; delay is in milliseconds. Training set: 06_03, 09_04, 10_01, 11_02; 27,451 training frames.

- Participant-independent generalization (Table 5; Figure 5)
    - Across unseen participants, the Causal LSTM retained the strongest mean accuracy and Macro F1, but participant-level variance remained substantial.
    - Probabilistic sequence models showed temporal smoothing benefits without matching the recurrent model's aggregate classification performance.

Table 5

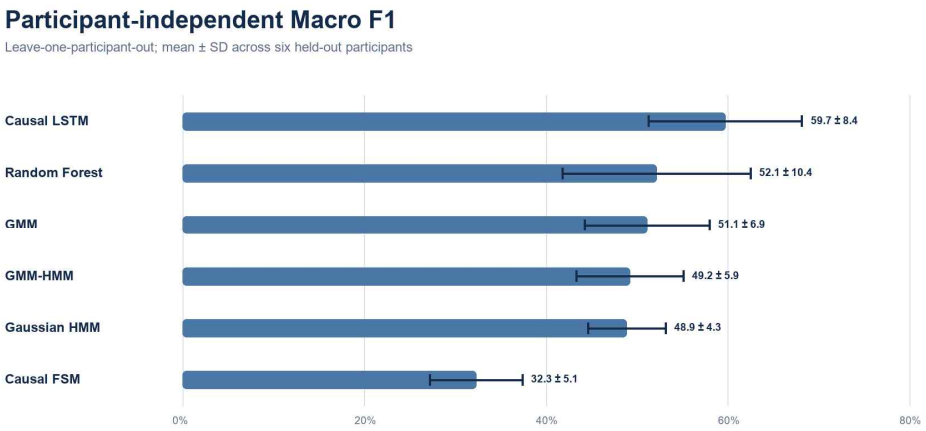
Leave-One-Participant-Out Performance

Model	±1 s Tol. Acc.	Full-seq. Macro F1	FREE F1	APP. F1	PAINT. F1	RET. F1	Trans. F1 (±1 s)
Causal FSM	67.2 (11.3)	32.3 (5.1)	35.9 (12.3)	9.5 (9.7)	74.9 (9.0)	8.9 (4.7)	4.2 (1.3)
Random Forest	84.4 (6.6)	52.1 (10.4)	68.9 (8.0)	31.7 (18.9)	85.6 (9.7)	22.4 (12.7)	15.4 (5.6)
GMM	79.0 (7.7)	51.1 (6.9)	67.7 (7.9)	32.3 (14.3)	83.4 (8.5)	21.1 (9.7)	8.8 (3.3)
Gaussian HMM	74.2 (8.6)	48.9 (4.3)	64.8 (9.0)	28.8 (7.1)	77.5 (11.7)	24.4 (9.6)	15.2 (2.0)
GMM-HMM	73.9 (9.8)	49.2 (5.9)	65.8 (8.4)	32.8 (14.0)	80.2 (11.5)	18.0 (6.8)	12.0 (5.2)
Causal LSTM	85.9 (5.5)	59.7 (8.4)	70.7 (8.1)	49.4 (14.8)	87.0 (7.4)	31.8 (12.6)	32.7 (12.8)

Note. Values are mean (SD) percentages across six participant folds. Tolerant accuracy permits ±1-s phase timing differences. Full-sequence Macro and phase F1 use strict framewise matching over every valid held-out frame. Transition F1 uses one-to-one matching of the same from-to event within ±1 s.

Figure 5

Participant-Independent Strict Full-Sequence Macro F1



Note. Bars show strict full-sequence Macro F1 means; error bars show one standard deviation across held-out participants.

VI. Discussion

- Model trade-offs
  - Causal FSM
    - Strength: transparent logic, negligible training demand, direct safety review
    - Weakness: brittle thresholds, poor transient-phase recognition, heavy calibration dependence
  - Random Forest
    - Strength: strong low-data baseline, nonlinear decision boundaries, simple fitting
    - Weakness: no temporal memory, greater susceptibility to framewise chattering and false transitions
  - GMM
    - Strength: high transition sensitivity, compact probabilistic phase representation
    - Weakness: phase-overlap ambiguity and many spurious transitions without temporal state
  - Gaussian HMM
    - Strength: causal temporal smoothing and interpretable state transitions
    - Weakness: rigid single-Gaussian emissions and delayed or weakened brief-phase responses
  - GMM-HMM
    - Strength: multimodal emissions with temporal continuity

        - Weakness: added complexity without top-ranked performance; dependence on learned transition statistics
    - Causal LSTM
        - Strength: strongest participant-independent aggregate performance; few false transitions
        - Weakness: greater data requirement, weaker interpretability, remaining transition-recall and delay trade-off
- Interpretation of training-structure and participant effects
    - Four-trial model ranking changed with participant and interface composition
    - Participant-level variance exceeded a simple single-trial comparison
    - Implication: model selection requires participant-independent evaluation rather than one held-out trial
- Limitations
    - Single wall geometry, EOAT, task family, and laboratory context
    - Only six participants; one common held-out trial for the training-structure comparisons
    - Severe class imbalance and short APPROACH/RETRACT intervals
    - Metric dependence on the selected ±1.0-s timing tolerance; tolerant accuracy and transition-event F1 must be interpreted alongside strict full-sequence F1
    - No closed-loop user evaluation of phase-triggered wall-normal shared control
- Next validation
    - Closed-loop shared-control experiment: safety, contact stability, operator workload, and task quality
    - Cross-geometry and cross-tool validation; uncertainty-aware assistance release

VII. Contributions

- Application-driven formulation of causal phase recognition for wall-normal shared control in remote robotic painting
- Common one-frame causal benchmark spanning rules, framewise learning, probabilistic sequence models, and recurrent learning
- Empirical evidence that four-trial composition changes model ranking and that participant-independent evaluation is required
- Evaluation emphasis on minority-phase F1, transition matching, false transitions, and delay beyond overall accuracy